

Xiaoteng Zhou

Underwater Acoustics | Scientific Computing | Physical Modelling & Experiments

zhou@robot.t.u-tokyo.ac.jp | ResearchGate: Xiaoteng Zhou | Based in Japan

PROFESSIONAL PROFILE

Applied scientist specialising in underwater acoustics and physical modelling, with a Ph.D. from the University of Tokyo and an M.E. in Ocean Engineering. Develops acoustic simulators, analytical and computational models, and Python-based algorithms for underwater sensing. Combines scientific computing with water-tank and field experiments, multisensor measurements and model validation, with applications in sonar perception, marine hydrodynamics, marine engineering and environmental sensing.

TECHNICAL SKILLS

Scientific computing	Python for acoustic simulation, experimental data processing and numerical analysis; C++; Robot Operating System (ROS).
Underwater acoustics	Acoustic propagation, reflection and backscatter; multipath and clutter modelling; acoustic measurements; High-resolution sonar imaging.
Acoustic data analysis	Underwater acoustic signal and sonar data processing; feature-based and AI-based methods for target detection, tracking and scene reconstruction.
Fluid mechanics	Hydrodynamic modelling and experimental analysis: drag, added mass, waves and currents; familiarity with computational fluid dynamics (CFD).
Sensing systems	Forward-looking sonar (FLS), side-scan sonar (SSS), synthetic aperture sonar (SAS), multibeam echosounders (MBES), acoustic Doppler current profiler (ADCP)
Experimental methods	Design and execution of recirculating water tank, towing-tank and field trials; acoustic measurements, multisensor integration, experimental validation and quantitative data analysis.

RESEARCH EXPERIENCE

The University of Tokyo <i>Project Researcher, Graduate School of Engineering (since Apr. 2026)</i> Doctoral research: Apr. 2023 – Mar. 2026. Contributions across both roles.	Apr. 2023 – Present
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- **Physical modelling and acoustic simulation.** Developed an acoustic simulator incorporating reflection, backscatter, multipath propagation and clutter models; developed and validated acoustic methods for 3-D scene reconstruction.
- **Experimental design and validation.** Designed and conducted water-tank and field experiments, integrated sonar with optical, hyperspectral and navigation sensors (IMU and DVL), and collected and analysed measurements for model validation and marine monitoring.
- **Hydrodynamics and measurements.** Conducted BlueROV2 towing-tank experiments to analyse drag and added mass; used sonar to measure waves and currents and investigate fluid phenomena.
- **Scientific software and data analysis.** Developed Python-based software for acoustic simulation, experimental data processing and numerical analysis; applied feature-based and AI-based methods to clutter mitigation, target detection, tracking and scene reconstruction.
- **Marine environmental projects.** Responsible for acoustic investigations covering marine pollutant surveys, plume monitoring, benthic environmental monitoring and ecological analysis, including pollutant abundance and depth-resolved density mapping.

University of Alberta <i>Visiting Researcher, Artificial Intelligence Laboratory</i> Developed physics-driven and AI-based acoustic processing algorithms for tracking underwater targets.	Mar. 2025 – Jul. 2025
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EDUCATION

The University of Tokyo <i>Ph.D. in Marine Acoustic Perception</i>	Apr. 2023 – Mar. 2026
Harbin Institute of Technology <i>M.E. in Ocean Engineering</i>	Sep. 2020 – Jul. 2022
Jimei University <i>B.E. in Navigational Technology</i>	Sep. 2016 – Jul. 2020

REPRESENTATIVE PUBLICATIONS

The representative works below highlight contributions to the derivation and analysis of acoustic principles, acoustic simulator development and applied underwater acoustics.

Journal Articles

- X. Zhou, Y. Wang, K. Mizuno, K. Tsutsumi, and H. Sugimoto, “An Automatic Sonar Target Shadow Extraction Approach for Rapid Seafloor 3-D Perception Using a Single Acoustic Camera”, *IEEE Transactions on Instrumentation and Measurement*, vol. 74, art. 9701018, 2025.
- X. Zhou, Y. Zhang, and K. Mizuno, “Systematic Acoustic Monitoring of Underwater Floating Debris Using Deep Learning With Temporal Information”, *IEEE Journal of Oceanic Engineering*, 2026.
- X. Zhou, M. Mandal, and K. Mizuno, “Flow-Guided Acoustic Imaging for Tracking Underwater Floating Debris in Low-Visibility Environments”, *Signal Processing*, vol. 251, art. 110896, 2027.
- X. Zhou and K. Mizuno, “Benchmarking AI-Driven Acoustic Monitoring for Floating Marine Debris: Challenges in Deep Learning-Based Debris Extraction”, *Marine Pollution Bulletin*, vol. 222, pt. 1, art. 118655, 2026.
- X. Zhou, Y. Wang, K. Mizuno, K. Tsutsumi, and H. Sugimoto, “High-Resolution Underwater Mapping in Low-Visibility and Confined Environments Using Imaging Sonar”, *Applied Ocean Research*, vol. 168, art. 104959, 2026.
- X. Zhou, K. Mizuno, Y. Zhang, K. Tsutsumi, and H. Sugimoto, “Acoustic Camera-Based Adaptive Mosaicking Framework for Underwater Structures Inspection in Complex Marine Environments”, *IEEE Journal of Oceanic Engineering*, vol. 49, no. 4, pp. 1549–1573, 2024.
- X. Zhou and K. Mizuno, “Acoustic Camera-Based Super-Resolution Reconstruction Approach for Underwater Perception in Low-Visibility Marine Environments”, *Applied Ocean Research*, vol. 150, art. 104110, 2024.
- X. Zhou, S. Yuan, C. Yu, H. Li, and X. Yuan, “Performance Comparison of Feature Detectors on Various Layers of Underwater Acoustic Imagery”, *Journal of Marine Science and Engineering*, vol. 10, no. 11, art. 1601, 2022.

Conference Contributions (OCEANS and ASA)

- X. Zhou, H. Tsuchiya, J. Ota, A. Yamashita, and Y. Wang, “Active Sonar Viewpoint Optimization via Reinforcement Learning for Marker-Based Underwater Navigation,” *2026 OCEANS Conference and Exposition*, Monterey, CA, USA, Sept. 2026. Accepted.
- X. Zhou and K. Mizuno, “Acoustic Camera-Based Automatic Abundance Estimation Algorithm for Marine Plastic Debris”, *IEEE/MTS OCEANS Halifax*, 2024. Student Poster Competition.
- X. Zhou, C. Yu, X. Yuan, Y. Wu, H. Feng, and C. Luo, “Nonlinear Intensity Sonar Image Matching Based on Deep Convolution Features,” *IEEE/MTS OCEANS Chennai*, 2022.
- X. Zhou, Y. Wang, and K. Mizuno, “Efficient and Robust 3-D Perception of Seafloor Targets Using a Single Forward-Looking Sonar for Low-Visibility Environments”, *Proceedings of Meetings on Acoustics*, vol. 59, art. 070004, 2025.
- X. Zhou, Y. Wang, and K. Mizuno, “Depth-Resolved Density Mapping of Suspended Marine Debris via High-Resolution FLS and Optical Flow”, *Journal of the Acoustical Society of America*, vol. 158, no. 4 suppl., p. A35, 2025 (meeting abstract).

Further research: Complete publication list available via the ResearchGate profile on page 1.

HONOURS, AWARDS AND MEMBERSHIP

- **SPRING GX**, The University of Tokyo; Japan Science and Technology Agency (JST): Fostering Advanced Human Resources to Lead Green Transformation Project.
- **Departmental Research Funding**, The University of Tokyo; one of two recipients.
- **CLIMATE X Leadership Training Program**, Global Alliance of Universities on Climate.
- **Member**, Acoustical Society of America (ASA).